

YIFAN JIANG

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EDUCATION

The University of Texas at Austin, Austin, USA

2020 – Present

Ph.D. in Electrical and Computer Engineering¹

Advisor: [Prof. Zhangyang \(Atlas\) Wang](#)

Research Interests: Computational Photography, Generative Model, 3D Vision, Efficient ML

Huazhong University of Science and Technology, Wuhan, China

2015 – 2019

B.E. in Electronic Information Engineering

PUBLICATIONS

(* indicates equal contribution)

1. **Y. Jiang***, D. Xu*, P. Wang, Z. Fan, H. Shi, and Z. Wang. “SinNeRF: Training Neural Radiance Fields on Complex Scenes from a Single Image”, European Conference on Computer Vision (**ECCV**) 2022.
2. **Y. Jiang***, Z. Fan*, P. Wang*, X. Gong, D. Xu, and Z. Wang. “Unified Implicit Neural Stylization”, European Conference on Computer Vision (**ECCV**) 2022.
3. **Y. Jiang**, B. Wronski, B. Mildenhall, J. T. Barron, Z. Wang, and T. Xue. “Fast and High-Quality Image Denoising via Malleable Convolutions”, European Conference on Computer Vision (**ECCV**) 2022.
4. **Y. Jiang**, X. Gong, Junru Wu, H. Shi, Z. Yan, and Z. Wang, “AutoX3D: Searching Ultra-Efficient Architecture for Video Understanding”, Winter Conference on Applications of Computer Vision (**WACV**) 2022.
5. **Y. Jiang**, S. Chang, and Z. Wang, “TransGAN: Two Pure Transformers can Make One Strong GAN and That Can Scale Up”, Advances in Neural Information Processing Systems (**NeurIPS**), 2021.
6. **Y. Jiang**, Z. He, J. Zhang, Y. Wang, Z. Lin, K. Sunkavalli, S. Chen, S. Amirghodsi, S. Kong, and Z. Wang, “SSH: A Self-supervised Framework for Image Harmonization”, International Conference on Computer Vision (**ICCV**), 2021.
7. **Y. Jiang**, X. Gong, D. Liu, Y. Cheng, C. Fang, X. Shen, J. Yang, P. Zhou, and Z. Wang, “EnlightenGAN: Deep Light Enhancement without Paired Supervision”, Transaction on Image Processing (**TIP**)
8. Z. Chen, **Y. Jiang**, D. Liu, and Z. Wang, “CERL: A Unified Optimization Framework for Light Enhancement with Realistic Noise”, Transaction on Image Processing (**TIP**)
9. D. Xu*, P. Wang*, **Y. Jiang**, Z. Fan, and Z. Wang, “Signal Processing for Implicit Neural Representations”, Advances in Neural Information Processing Systems (**NeurIPS**), 2022.
10. D. Xu, H. Poghosyan, S. Navasardyan, **Y. Jiang**, H. Shi, and Z. Wang, “ReCoRo: Region-Controllable Robust Light Enhancement by User-Specified Imprecise Masks”, ACM Multimedia (**MM**), 2022
11. B. Pan, R. Panda, **Y. Jiang**, Z. Wang, R. Feris, and A. Oliva, “IA-RED2: Interpretability Aware Redundancy Reduction for Vision Transformers”, Advances in Neural Information Processing Systems (**NeurIPS**), 2021.
12. Y. Fu, Z. Yu, Y. Zhang, **Y. Jiang**, C. Li, Y. Liang, M. Jiang, Z. Wang, and Y. Lin, “InstantNet: Automated Generation and Deployment of Instantaneously Switchable Precision Networks”, Design Automation Conference (**DAC**), 2021.
13. T. Meng*, X. Chen*, **Y. Jiang**, and Z. Wang, “A Design Space Study for LISTA and Beyond”, International Conference on Learning Representations (**ICLR**), 2021.
14. X. Gong, S. Chang, **Y. Jiang**, and Z. Wang. “AutoGAN: Neural Architecture Search for Generative Adversarial Networks”, International Conference on Computer Vision (**ICCV**), 2019.

IN-SUBMISSION

1. Q. Wu, X. Chen, **Y. Jiang**, and Z. Wang, “Chasing Better Deep Image Priors Between Over- and Under-parameterization”, International Conference on Learning Representation (**ICLR**), 2023, In Submission.
2. Z. Fan, P. Wang, **Y. Jiang**, X. Gong, D. Xu, and Z. Wang, “NeRF-SOS: Any-View Self-supervised Object Segmentation from Complex Real-World Scenes”, International Conference on Learning Representation (**ICLR**), 2023, In Submission.

¹Studied at Texas A&M University from Aug. 2019 to Aug. 2020; then transferred with my advisor to UT Austin

3. **Y. Jiang**, P. Hedman, B. Mildenhall, D. Xu, J. T. Baron, Z. Wang, and T. Xue, “AlignNeRF: Improved High-fidelity Neural Radiance Field via Alignment-aware Training”, Computer Vision and Pattern Recognition (**CVPR**), 2023, In Submission.
4. **Y. Jiang**, Z. Xia, Z. Fan, C. Zhang, X. Zhang, and J. Chen, “Lidar-guided Video Depth Estimation on Smartphones”, Computer Vision and Pattern Recognition (**CVPR**), 2023, In Submission.
5. D. Xu, **Y. Jiang**, W. Cong, P. Wang, Z. Fan, Y. Wang, and Z. Wang, “3D-Diffusion: Toward 3D Consistent View Generation via Diffusion Model”, Computer Vision and Pattern Recognition (**CVPR**), 2023, In Submission.
6. Q. Wu, **Y. Jiang**, J. Wu, K. Wang, G. Zhang, H. Shi, Z. Wang, and S. Chang. “Grasping the Arrow of Time from the Singularity: Decoding Micromotion in Low-dimensional Latent Spaces from StyleGAN”, Transaction on Image Processing, (**TIP**), In Submission.

INTERNSHIP EXPERIENCE

Adobe, San Jose, USA

May. 2022 – Present

Research Intern with [Marc Levoy's Team](#), Adviser: [Dr. Zhihao Xia](#), [Dr. Cecilia Zhang](#), [Dr. Jiawen Chen](#).

- Working on monocular video depth estimation

Google Research, Mountain View, USA

May. 2021 – May. 2022

Research Intern with [GCam](#), Adviser: [Dr. Tianfan Xue](#), [Bart Wronski](#), [Dr. Ben Mildenhall](#), [Dr. Jon Barron](#).

- Developed a fast denoising network by predicting spatially-varying kernels at low resolution and using a fast fused op to jointly upsample and apply these kernels at full resolution. The resultant paper was accepted by ECCV'2022
- Designed a high-fidelity neural radiance field that can render high-quality novel view images.

Adobe, San Jose, USA

May. 2020 – Nov. 2020

Research Intern with Applied Research Team (ART), Adviser: [Dr. He Zhang](#) and [Dr. Jianming Zhang](#).

- Developed a self-supervised method for image harmonization that does not require human annotation labels. The resultant paper was accepted by ICCV'2021

Bytedance AI Lab, Beijing, China

Jan. 2019 – Aug. 2019

Research Intern with US CV Lab, Adviser: [Dr. Jianchao Yang](#) and [Dr. Xiaohui Shen](#) and [Dr. Ding Liu](#).

- Designed a jointly image denoising and low-light enhancement algorithm, which was integrated in the selfie camera app [FaceU](#).

MEDIA HIGHLIGHT

- TransGAN was covered by [Quanta Magazine](#) (Mar. 2022) and was highlighted by [Top AI influencers](#) and high-profile [YouTubers](#), as well as considered as [the most influential new paper of the month \(Feb. 2021\)](#).
- AutoGAN was covered by [Synced AI Technology Industry Review](#) (Aug. 2019), and also featured on [Towards Data Science](#) (Sep. 2019) and [Analytics Magazine](#) (Aug. 2019), etc.

COMMUNITY SERVICES

- Reviewer for: CVPR'2021-2022, ICCV'2021, ECCV'2022, ICML'2022, NeurIPS'2022, ICLR'2023, Siggraph Asia'2022, Siggraph'2022, WACV'2022, Transaction on Image Processing (TIP), International Journal of Computer Vision (IJCV), NeuroComputing, IEEE Robotics and Automation Letters (RA-L)
- Workshop Organizer for: [ECCV RLQ-TOD Workshop 2020](#)

INVITED TALKS

- “Fast and High-Quality Image Denoising via Malleable Convolutions” at Adobe, Marc Levoy's team.
- “Vision Transformer for Image Generation, Editing, and Processing” at Google Research, Computational Photography Team.
- “TransGAN: Two Transformers Can Make One Strong GAN” at [\[cai-workshop\]](#), [\[SHI Lab @University of Oregon\]](#)